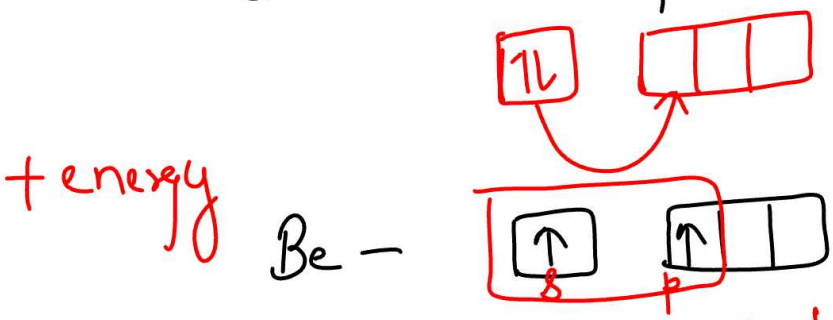
Basic chemistry →

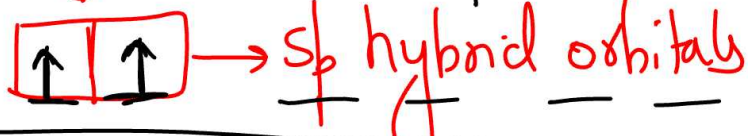
Hybridisation → It is the intermixing of valence orbitals of central atom to get new orbitals which have same shape & energies (hybrid orbitals).

$BeCl_2$ → Be

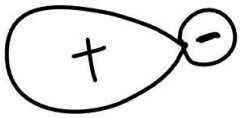
Be – $1s^2\ 2s^2\ 2p^0$



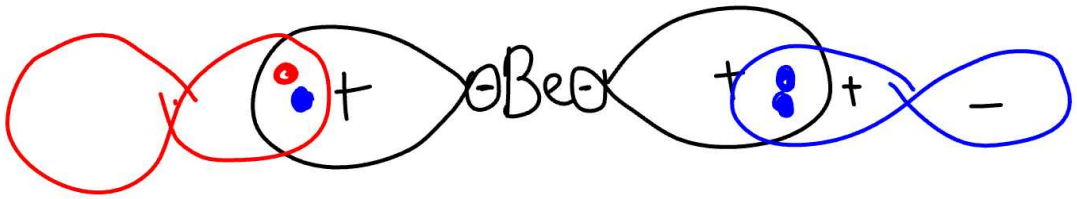
sp hybridisation



In general, shape of hybrid orbital →



Geometry = linear



or Cl — Be — Cl Bond angle = 180°

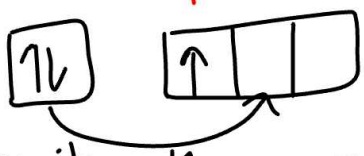
180°

Cl – $3s^2 3p^5$

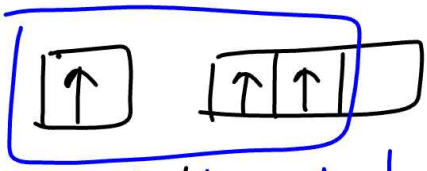
Cl – $3s^2 3p^5$

2. BF_3

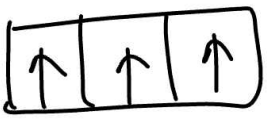
B – $2s^2\ 2p^1$



+ energy to excite the e⁻

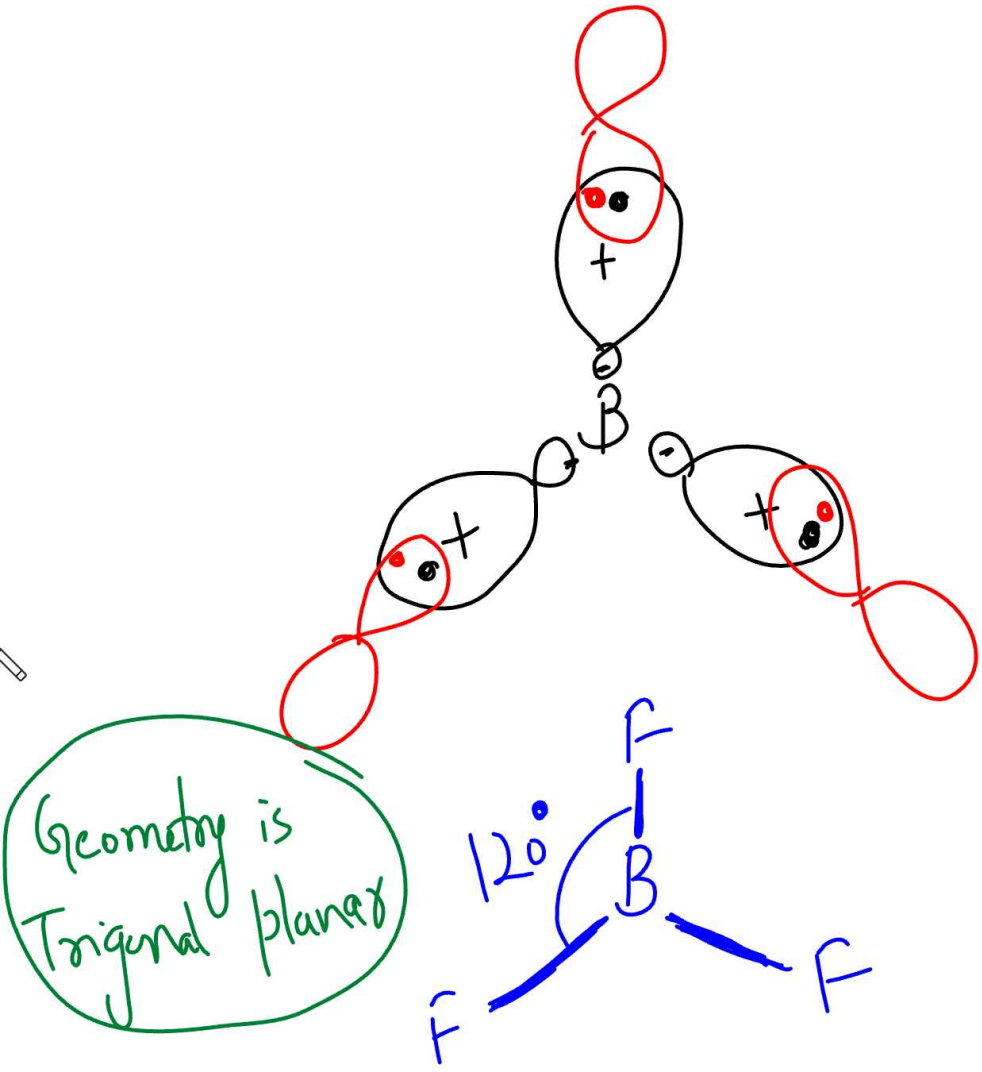


sp^2 hybridisation

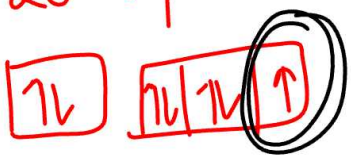


→ sp^2 hybrid orbitals

oriented at 120°



F - $2s^2$ $2p^5$

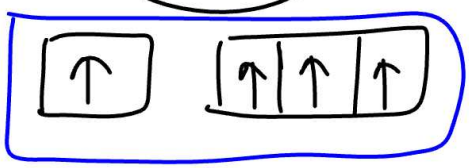


CH_4

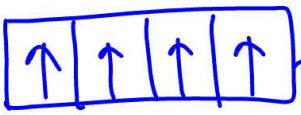
C – $2s^2$ $2p^2$



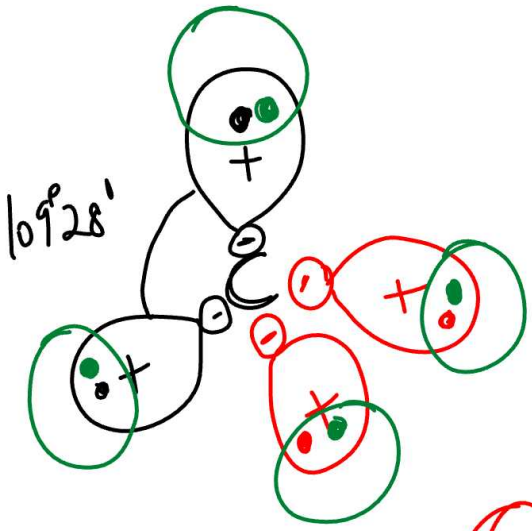
C –



sp^3 hybridisation

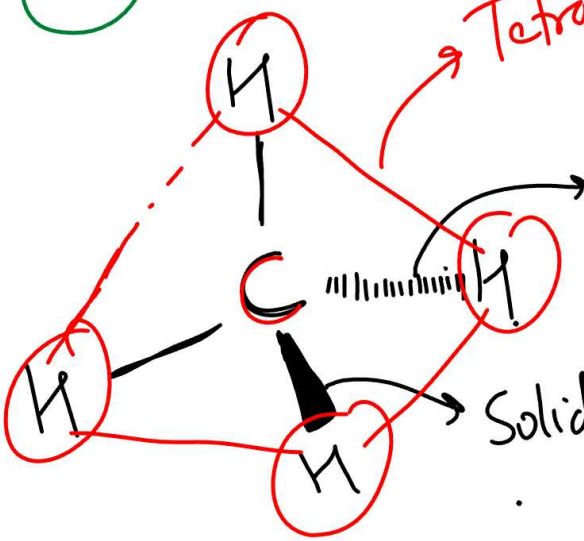


→ oriented at angle of $109°28'$ ⇒ tetrahedral



4H – 1s¹





→ Tetrahedral

dashed wedge → bond away from you.

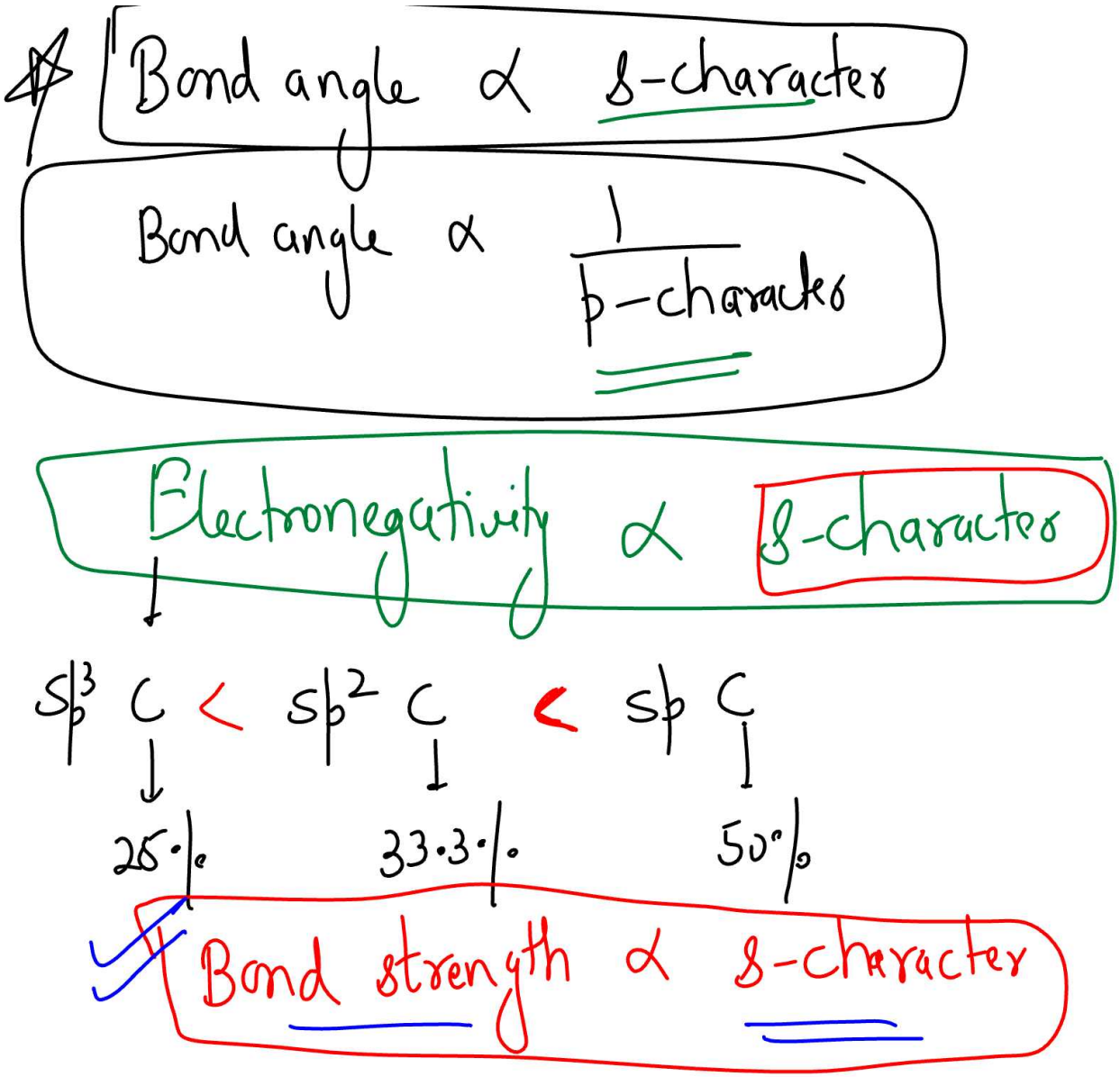
Solid wedge → represents a bond pointing towards an observer.

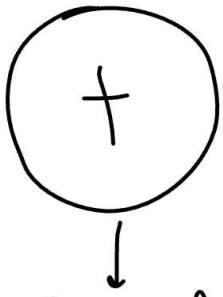
Type of hyb.	Geometry	Bond angle	% s-ch.
sp	Linear	180°	50% ($\frac{1}{2}$) (50% p)
sp^2	Trigonal planar	120°	33.33 ($\frac{1}{3}$) (66.67% p)
sp^3	Tetrahedral	109°28'	$\frac{1}{4} \times 100$ (25%) (75% p)

$$\%\ s\text{-ch} = \frac{\text{no. of s orbitals}}{\text{total orb}} \times 100$$

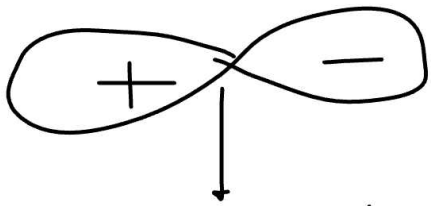
1 apple + 2 bananas → 🍸

half apple + half banana


Bond angle α s-character
Bond angle α 1 / p-character
Electronegativity α s-character
sp3 C < sp2 C < sp C
25%
33.3%
50%
Bond strength α s-character

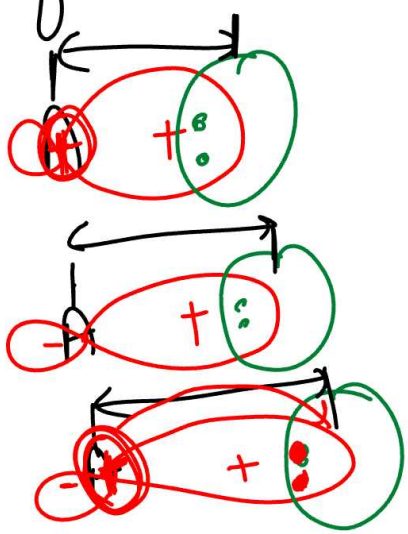


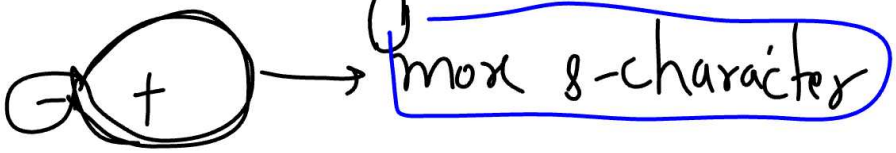
Short & bulky

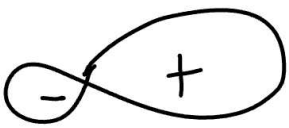


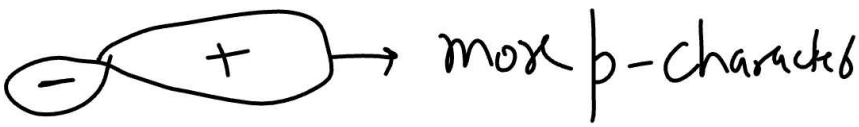
thin & longer

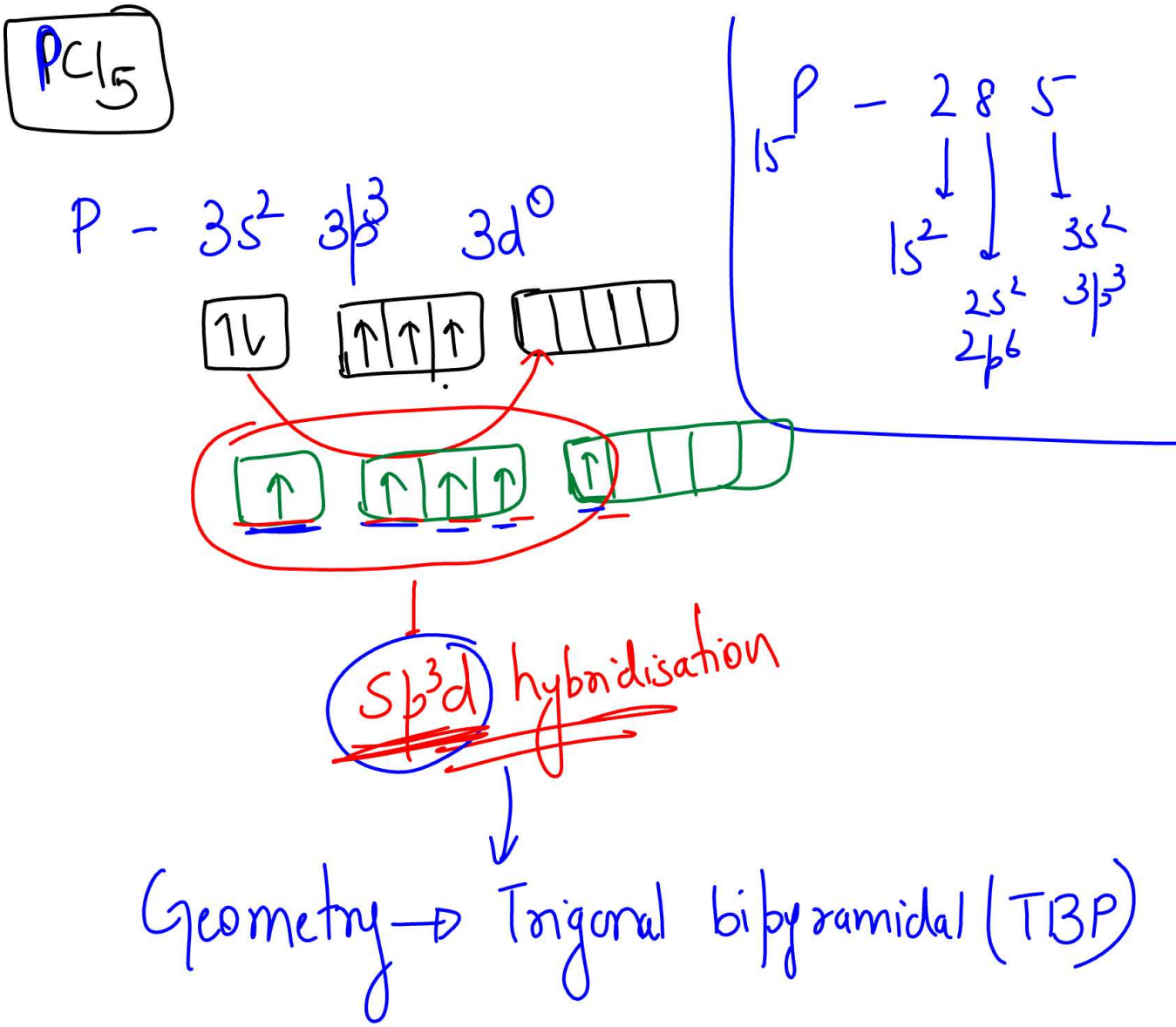
If s-character is increasing →

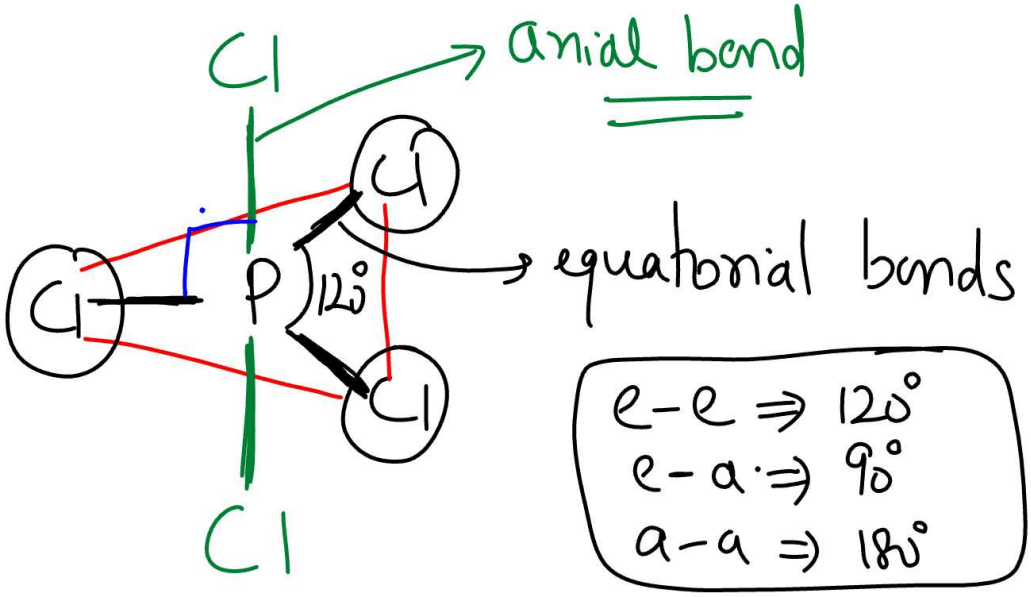









PCl5
P - 3s2 3p3 3d0
P - 2 8 5
1s2
2s2 2p6
3s2 3p3
sp3d hybridisation
Geometry → Trigonal bipyramidal (TBP)



e-e ⇒ 120°
e-a ⇒ 90°
a-a ⇒ 180°

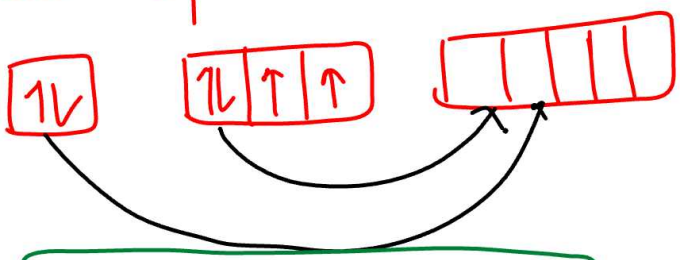
sp^3d hybridisation → geometry is TBP

sp^3d^2 eg: SF_6

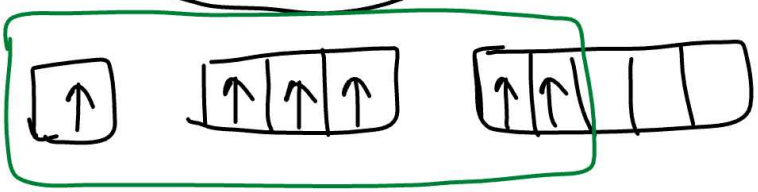
$_{16}S - 2\ 8\ 6$

I II 3rd shell

S - $3s^2$ $3p^4$ $3d^0$

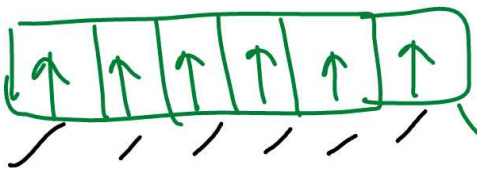


S^* -

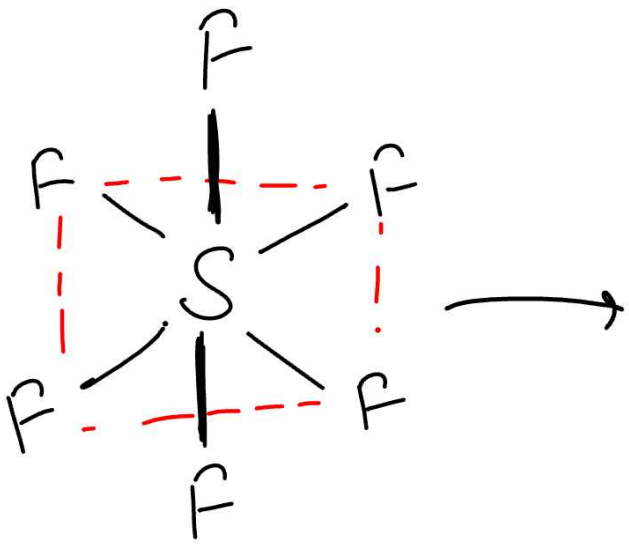


sp^3d^2 hybridisation

Oriented at 90°



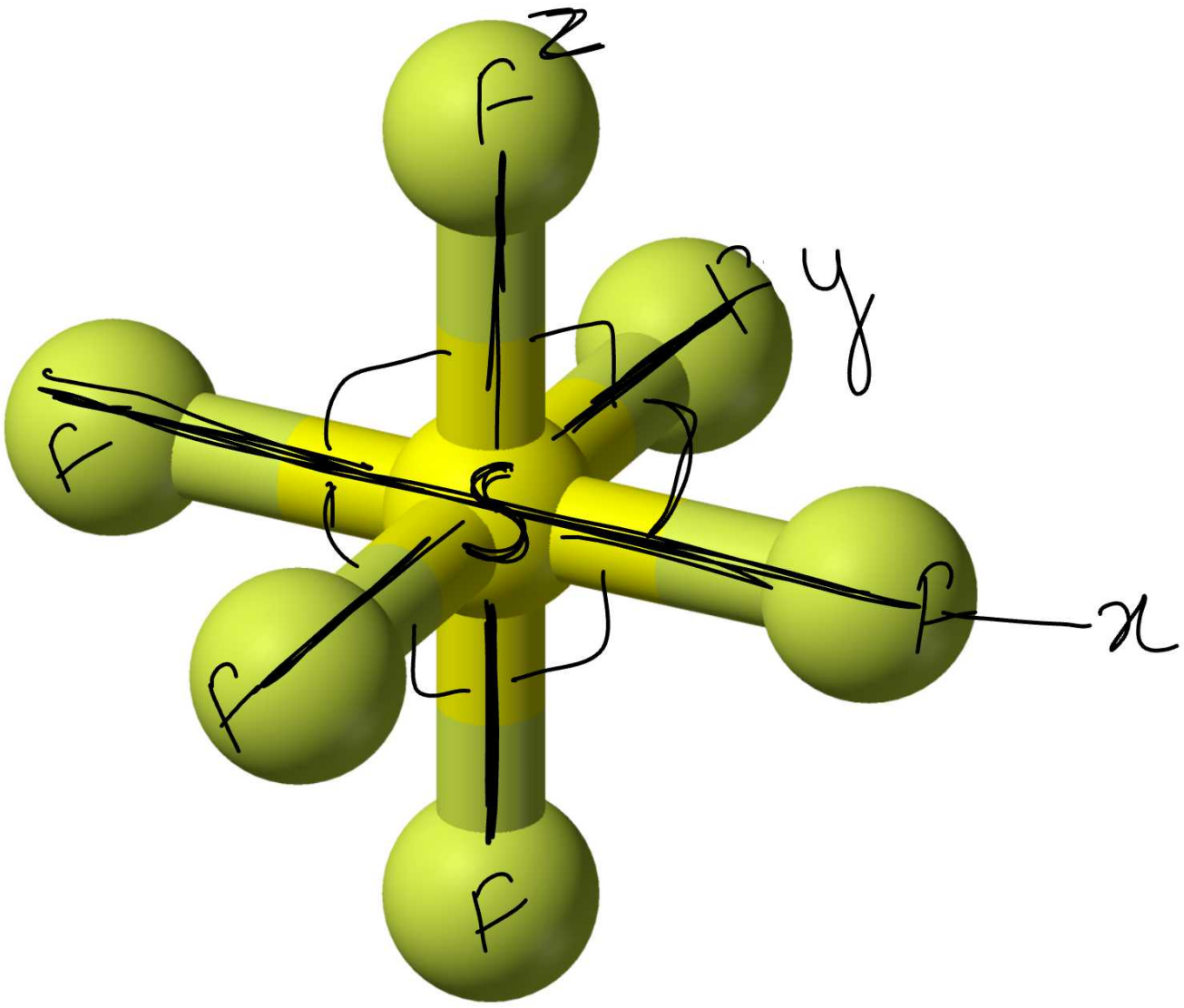
Octahedral

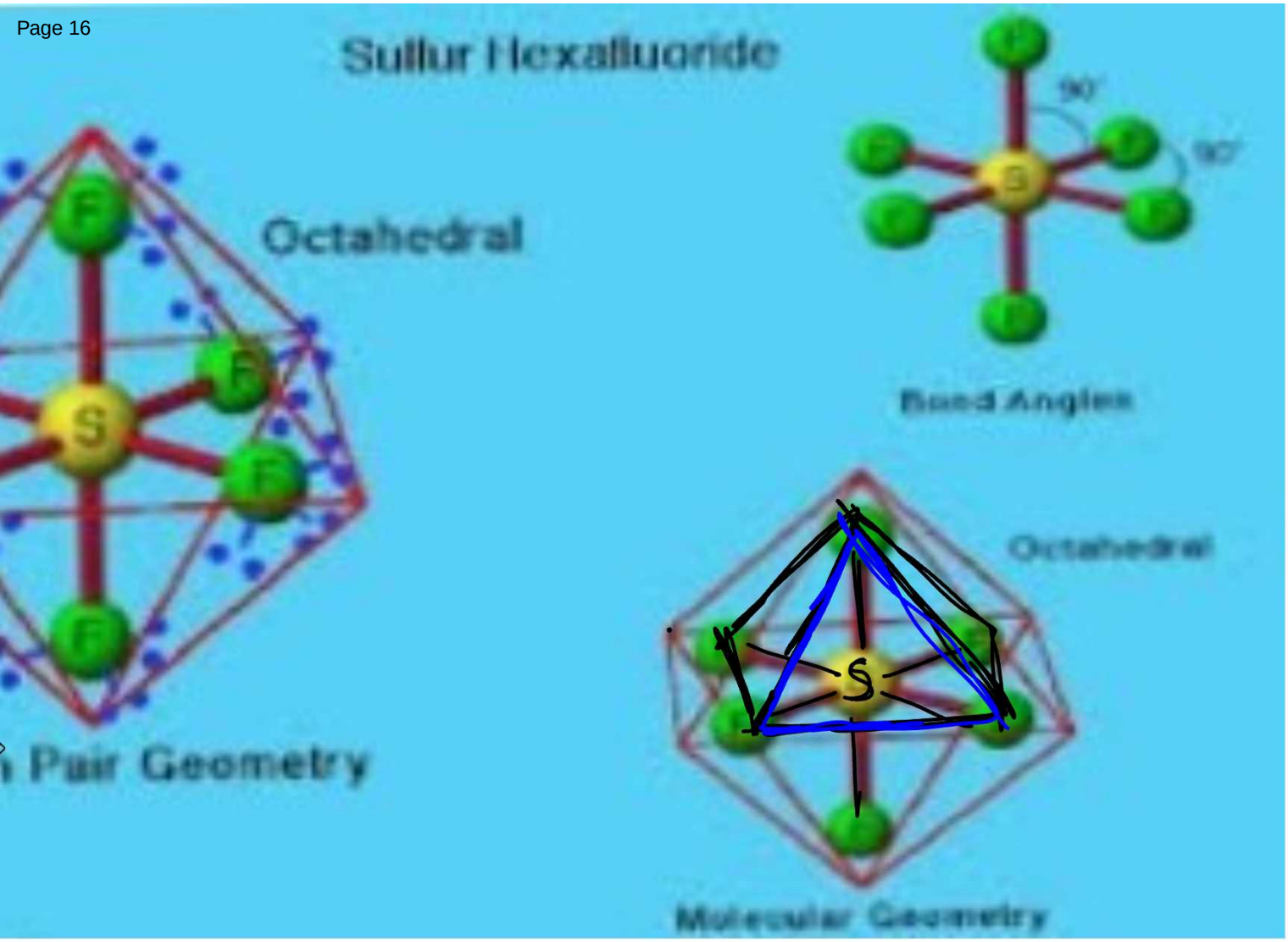


8 triangular faces

Octahedral geometry

In oct. geometry, bonds are formed along the x, y & z-axes


z
y
x


Sulfur Hexafluoride
Octahedral
Pair Geometry
90°
90°
Bond Angles
Octahedral
Molecular Geometry

Q1. Which of these have tetrahedral geometry.

a) CCl_4 → sp^3 → tetrahedral

b) PCl_5^- → sp^3d

c) PBr_5^- → sp^3d

d) SF_6 → Oct

Which of these is not linear.!

a) CO_2 → linear

b) BeF_2 → linear

c) BF_3 → sp^2 hybrid → Trigonal

d) O_2 $O=O$ → linear

σ-bond	π-bond
→ formed by head on overlap	→ formed by lateral overlap
→ Stronger	→ weaker
→ Can exist alone.	→ always exists with σ-bonds.
→ formed by hybrid orbitals.	→ They are formed by unhybrid p or d-orbitals.
→ σ-bond decide the geometry of molecule	→ π-bonds do have no role in deciding geometry.

How to count σ & π-bond

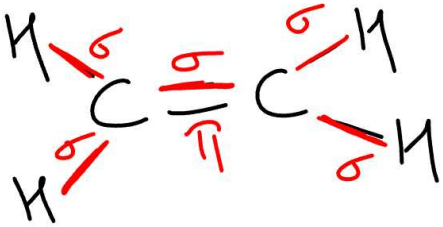
σ-bond is formed first & it is one only b/w two atoms.

Single bond → σ bond

↳ double bond → = → 1σ & 1π

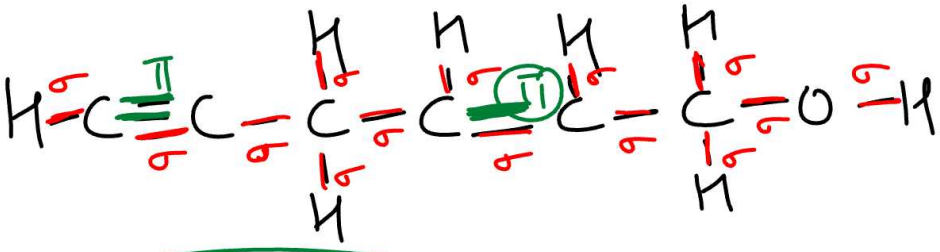
↳ Triple bond → ≡ → 1σ & 2π

$CH_2 = CH_2$



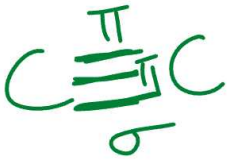
σ bonds = 5

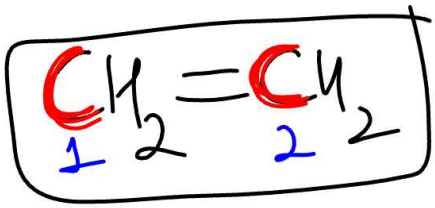
π-bonds = 1

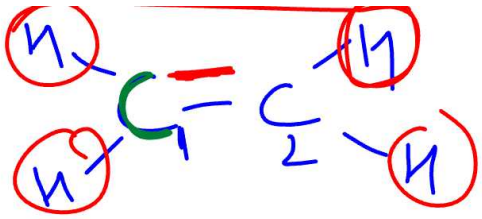


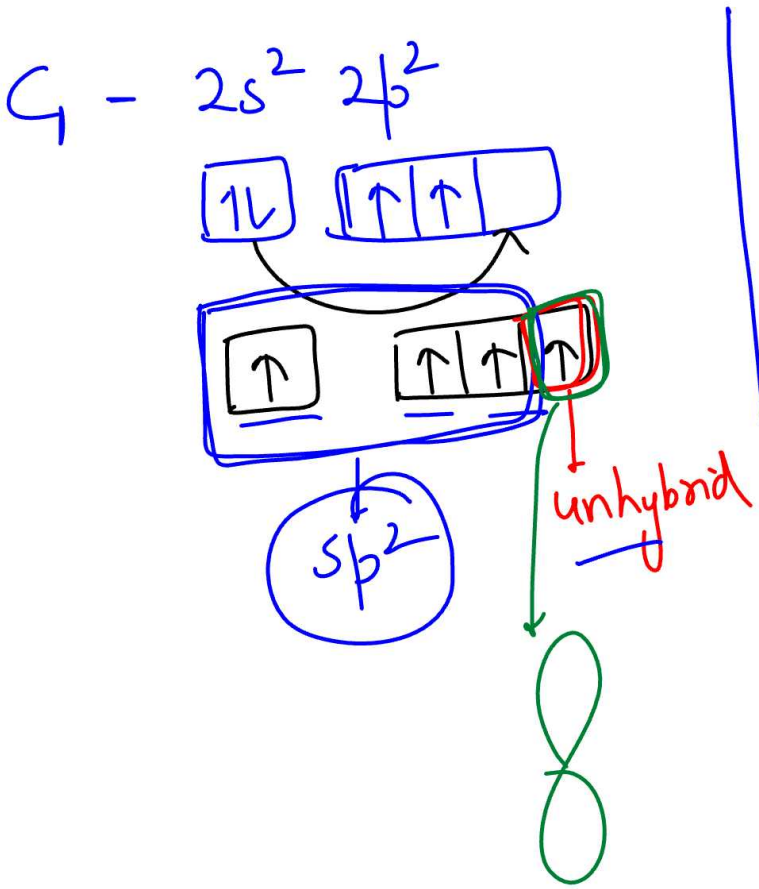
σ-bonds = 14

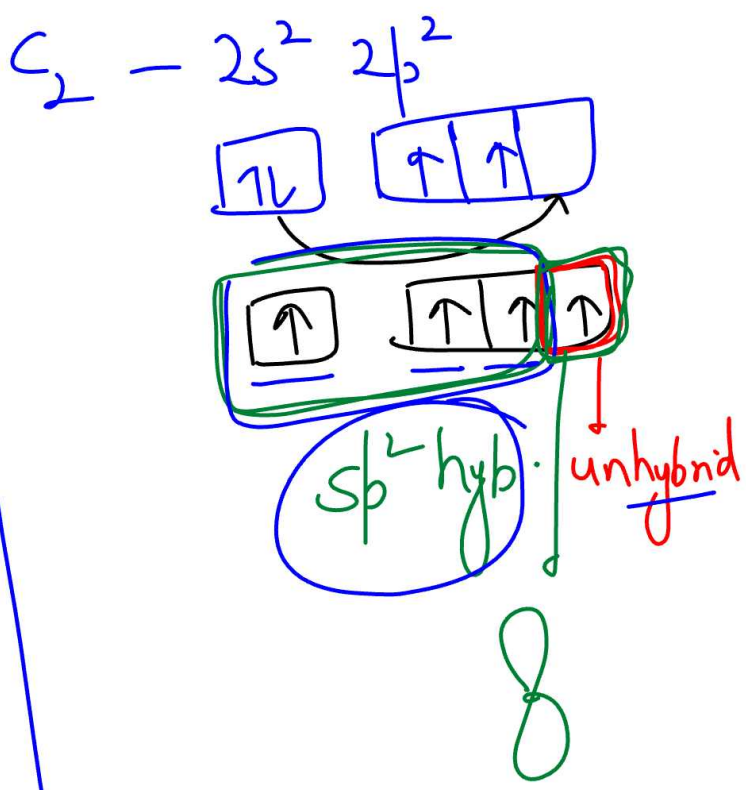
π bonds = 3

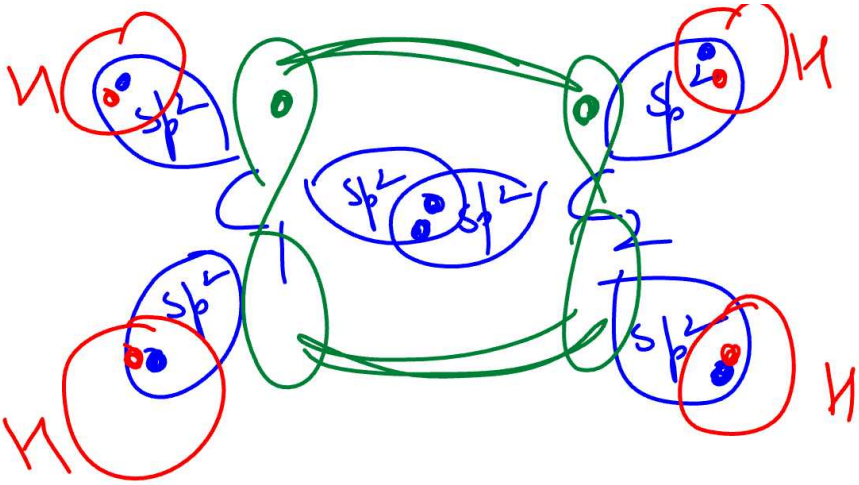



CH2=CH2
1 2 2

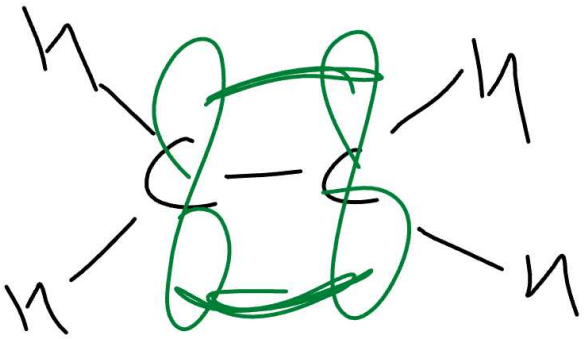

H
H
H
H
C1
C2


C1 - 2s2 2p2
sp2
unhybrid

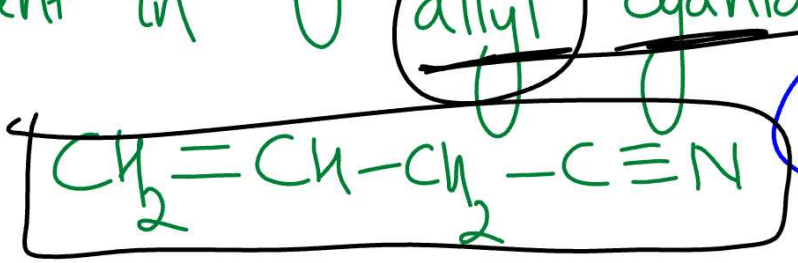

C2 - 2s2 2p2
sp2 hyb.
unhybrid



H – ls



How many σ & π-bonds are present in allyl cyanide



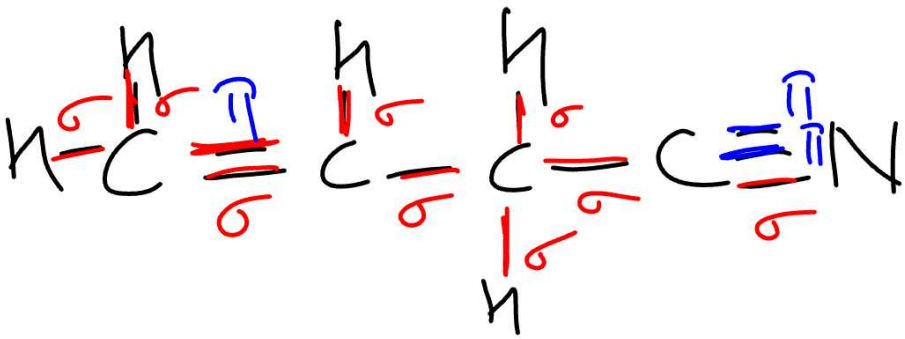
JEE advanced

a) 8σ, 2π

b) 9σ, 3π

c) 10σ, 3π

d) 9σ, 2π



9σ & 3π

Revise & red hybridisation part from NCERT - 11